



The QE65 Pro is our next-generation scientific-grade spectrometer with high sensitivity and low stray light performance that make it ideal for a wide range of low light level applications such as fluorescence, DNA sequencing and Raman analysis.

The enhanced QE65 Pro has a thermally stable design promoting excellent spectrometer wavelength stability, triggering functions for providing accurate timing and synchronization between the spectrometer and other devices and a replaceable slit design for added convenience. An extensive range of gratings offers flexibility in optimizing applications.

Robust Optical Bench

The QE65 Pro is the result of feedback from customers like you who appreciate high performance at a reasonable price and appreciate the flexibility of our modular approach to miniature spectroscopy. Consider some of the enhancements we've made to the QE65 Pro:

- Availability of new gratings, selected for their optimal wavelength range and stray light performance characteristics
- New mirror options to tailor reflectivity to your wavelength range
- Rectangular mirrors for more sensitivity at the ends of the spectral range
- SMA 905 connector assembly with replaceable slit design for added convenience and excellent repeatability
- Changes to the optical bench and its components that promote excellent spectrometer wavelength thermal stability

QE65 Pro

Scientific-Grade Spectrometer

Physical	
Dimensions:	182 mm x 110 mm x 47 mm
Weight:	1.18 kg (without power supply)
Detector	
Detector:	Hamamatsu S7031-1006, low etaloning
Detector range:	200-1100 nm
Pixels:	1024 x 58 (1044 x 64 total pixels)
Pixel size:	24 μm^2
Pixel well depth:	300,000 electrons/well, 1.5 m elec/column
Sensitivity:	22 electrons/count all λ ; 26 photons/count @ 250 nm
Quantum efficiency:	90% peak; 65% at 250 nm
Optical Bench	
Design:	f/4, Symmetrical crossed Czerny-Turner
Focal length:	101.6 mm input and output
Entrance aperture:	Field replaceable 5, 10, 10S, 25, 50, 100 or 200 μm wide
Grating options:	Multiple grating options, UV through Shortwave NIR
HC-1 grating:	Provides 200-1050 nm range (best efficiency)
OFLV filter options:	OFLV-QE (200-950 nm); OFLV-QE-250 (250-1000 nm); OFLV-QE-300 (300-1050 nm); OFLV-QE-350 (350-1100 nm); OFLV-QE-400 (400-1150 nm)
Other filter options:	Longpass OF-1 filters
Fiber optic connector:	SMA 905 to 0.22 NA single-strand optical fiber
Thermal stability:	<3.0 pixel shift from -10 to +50 $^{\circ}\text{C}$
Spectroscopic	
Wavelength range:	Grating dependent
Optical resolution:	~0.14-7.7 nm (FWHM)
Signal-to-noise ratio:	1000:1 (at full signal)
A/D resolution:	16 bit
Dark noise:	3 RMS counts
Dynamic range:	7.5×10^9 (system), 25000:1 for a single acquisition
Integration time:	8 ms-15 minutes
Stray light:	<0.08% at 600 nm; 0.4% at 435 nm
Corrected linearity:	>99%
Electronics	
Power consumption:	500 mA @ 5 VDC (no TE cooling); 3.5 A @ 5 VDC (with TE cooling)
Data transfer speed:	Full scans to memory every 8 ms with USB 2.0 port, 18 ms with USB 1.1 port, 300 ms with serial port
Inputs/Outputs:	10 onboard digital user-programmable GPIOs (general purpose inputs/outputs)
Breakout box:	Yes, HR4-BREAKOUT
Trigger modes:	4 modes
Gated delay feature:	Yes
Connector:	30-pin connector
Power-up time:	<5 seconds

Back-thinned Detector

At the heart of the QE65 Pro is a Hamamatsu FFT-CCD detector distinguished by its high quantum efficiency (90% maximum) and low etalon characteristics. Unlike linear CCDs, the 2D area detector in the QE65 Pro allows us to bin (or sum) a vertical row of pixels, which offers significant improvement in the signal-to-noise ratio (>1000:1) and signal processing speed of the detector.

What's more, the TE-cooled (down to -15 °C) detector has low noise and low dark signal, which enables low-light-level detection and long integration times – from 8 milliseconds to 15 minutes, with virtually no spectral distortion.

Ideal For Raman

QE65 Pro is a great choice for fluorescence and low light level applications and is particularly well suited for modular Raman spectroscopy.



In this setup, the QE65 Pro anchors a modular, probe-based Raman system. You'll note a bifurcated optical probe with one leg coupled to a 785 nm laser (in middle of photo) and the other to the spectrometer.

Optimize Your QE65 Pro

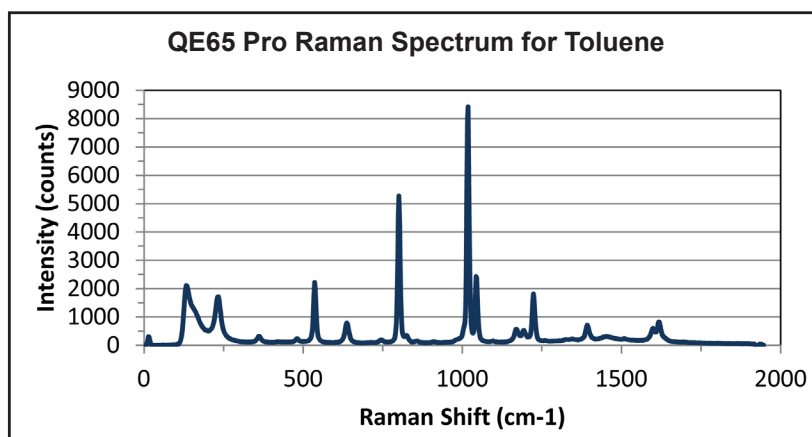
The QE65 Pro can be configured for a wide range of applications thanks to the availability of multiple gratings and optical bench accessories. We've added grating options that provide added flexibility for UV applications, Raman analysis (see item below) and more. The table below describes our newest gratings. Efficiency curves and other information are available at oceanoptics.com.

When selecting gratings, remember that your choice often involves trade-offs. For example, gratings with very high groove density (mm⁻¹) allow greater optical resolution but at the expense of a truncated spectral range. Spectral range is also affected by the starting wavelength of the grating. Please consult an Applications Scientist for the best grating for your application.

Grating Number	Intended Use	Groove Density	Blaze Wavelength	Best Efficiency (>30%)
H33	Absorbance	300	200 nm	200-450 nm
H34	UV Raman	3600	250 nm	200-550 nm
H35	NIR Raman	1200	1000 nm	525-1625 nm
H36	VIS Raman	900	500 nm	325-1225 nm
H5U	Absorbance	1200	250 nm	225-450 nm
H7U	Absorbance	2400	240 nm	200-800 nm
H10U	Absorbance	1800	250 nm	210-475 nm

Best Choice for Raman

For Raman spectroscopy, we offer a 900 mm⁻¹ grating (#H36) that is blazed at 500 nm and has good response at the 532 nm and 785 nm laser excitation wavelengths typical of Raman spectroscopy. Add lasers, sampling devices, software and more to build a complete Raman system, with the flexibility to make a variety of low light level measurements.



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